ChatGPT

Selected Research Papers and Insights

Paper (Reference)	Topic / Technique	Publication / Date	PDF	Relevance to IntelliShield	Subjects / Implementation
BertGCN (Alsalmi *et al.*, 2025) 【48†L1-L9】	Contextual BERT embedding + GCN on raw logs (no parser) 【48†L1-L9】	*Computers, Materials & Continua* 86(2), Dec 2025	PDF	Combines semantic (BERT) and structural (GCN) analysis of logs【48†L1-L9】, improving anomaly detection beyond fixed rules. Operates directly on raw text with high F1 on HDFS/BGL.	**Python:** PyTorch Geometric (GNN), NLP (BERT). **DBMS:** store graph data (Neo4j/RDF). **Networks:** system log events. **Stats:** classification metrics (precision/recall).
OntoLogX (Cotti *et al.*, 2026) 【49†L22-L31】	LLM (RAG) + log ontology to build knowledge graph; MITRE ATT&CK mapping 【49†L22-L31】	ArXiv, Apr 2026	PDF	LLMs + ontology transform raw logs into structured KGs, then aggregate sessions and predict MITRE tactics【49†L22-L31】. Delivers actionable threat intelligence (linking logs to attack steps).	**Python:** LLM APIs (e.g. HuggingFace), RDF/OWL libs. **DBMS:** graph/triple store (SPARQL queries). **Networks:** embeds security semantics (attacks). **Stats:** evaluate KG completeness.

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LogCraft (Zhang *et al.*, 2024) 【37†L47-L55】	End-to-end AutoML for unsupervised log anomaly detection 【37†L47-L55】	ASE '24 (Oct 2024)	PDF	Automates feature engineering and model selection for log anomaly detection 【37†L47-L55】, achieving F1≈0.899 without manual tuning. Suggests using AutoML (e.g. Auto-sklearn, TPOT) to streamline pipeline.	**Python:** AutoML tools, pandas; **DBMS:** log storage, metadata; **Networks:** multiple log sources; **Stats:** unsupervised methods, evaluation metrics.
LogLLM (Guan *et al.*, 2024) 【69†L17-L25】 【69†L26-L30】	BERT for semantic vectors + LLaMA decoder for classification; aligns embeddings; no log parser 【69†L17-L25】 【69†L26-L30】	ArXiv, Apr 2025	PDF	Uses BERT to encode log message semantics and LLaMA (decoder) to classify sequences, aligning their spaces. Eliminates template parsing (regex-prep) and outperforms prior SOTA on benchmarks 【69†L17-L25】 【69†L26-L30】.	**Python:** Transformers (HuggingFace BERT/ LLaMA), data processing; **DBMS:** raw log database; **Networks:** log datasets (HDFS, Hadoop); **Stats:** training strategy (three-stage).

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GPT-Log (Yin *et al.*, 2025) 【59†L17-L25】 【59†L32-L41】	GPT-2 fine-tuned on event-ID log sequences; Focal Loss for class imbalance 【59†L17-L25】 【59†L32-L41】	ArXiv, Oct 2025	PDF	Parses raw logs to event-ID sequences (Drain parser), then fine-tunes GPT-2 with focal loss to handle rare anomalies 【59†L17-L25】 【59†L32-L41】. Optimized GPT-2 significantly improves precision/recall, matching or beating GPT-3.5.	**Python:** HuggingFace Transformers (GPT-2), tokenization; **DBMS:** log and template storage; **Networks:** system logs; **Stats:** focal loss design, metrics (F1).
Unstable Log GPT (Hadadi *et al.*, 2024) 【66†L54-L61】	Fine-tuned GPT-3 on evolving (unstable) logs 【66†L54-L61】	ArXiv, 2024	PDF	Empirical study: a fine-tuned LLM (GPT-3) slightly outperforms supervised baselines on logs from software evolution 【66†L54-L61】. Shows LLM adaptability to log format changes.	**Python:** LLM fine-tuning (OpenAI API or similar), data versioning; **DBMS:** maintain historical logs; **Networks:** logs from multi-version systems; **Stats:** comparative analysis.
KnowGraph (Zhou *et al.*, 2024) 【71†L29-L38】	GNN anomaly detection + probabilistic logic (knowledge rules) 【71†L29-L38】	CCS '24 (Oct 2024)	PDF	Integrates domain knowledge (weighted first-order rules) into a GNN-based detector 【71†L29-L38】. Improves generalization and precision under distribution shifts. Suggests adding expert rules (e.g., intrusion patterns).	**Python:** graph ML libraries, MLN or Markov logic libs; **DBMS:** rule repository (perhaps RDF); **Networks:** model network flows as graphs; **Stats:** probabilistic inference (MLN), PCS analysis.

Each **Relevance** entry above cites key findings from the referenced paper (as specified by the 【...】 citations) and highlights how its approach could enhance IntelliShield's log-audit capabilities by applying Python-based ML/AI techniques, DB design, networking insights, and statistical methods from the listed courses. Each **Subjects/Implementation** cell maps the technique to Python, DBMS, Networks, and Statistics knowledge as requested.

Sources: Key results and methods are drawn from the cited papers【48†L1-L9】【49†L22-L31】【37†L47-L55】【69†L17-L25】【59†L17-L25】【66†L54-L61】【71†L29-L38】. Each citation corresponds to the given connected source content.